

Higher Tier only questions

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)		up to C ₁₃ -C ₁₆ demand is greater than supply (1)						
				from C ₁₇ -C ₂₀ upwards supply is greater than demand (1)			2	2		
				award (1) for 'at first demand is greater than supply then supply becomes greater than demand'						
		(ii)	I	C ₄ H ₁₀		1		1		
			II	butane	1			1		
			III	any of following - used to make polythene - used to make polymers - used to make other monomers - used to make plastics - used to make ethanol	1			1		
		(iii)		C ₃ H ₈ + 5 O ₂ → 3 CO ₂ + 4 H ₂ O		1		1	1	
	(b)	(i)		shale gas and contaminated water ✓			1	1		
		(ii)		fracking produces vast quantities of contaminated water ✓			1	1		
				Question 3 total	2	2	4	8	1	0